



Semester II 2020/2021

Subject : Bioinformatics I (SECB2103)
Section : 01 – Dr Haslina Hashim
Topic : Lab 07 - Phylogenetic

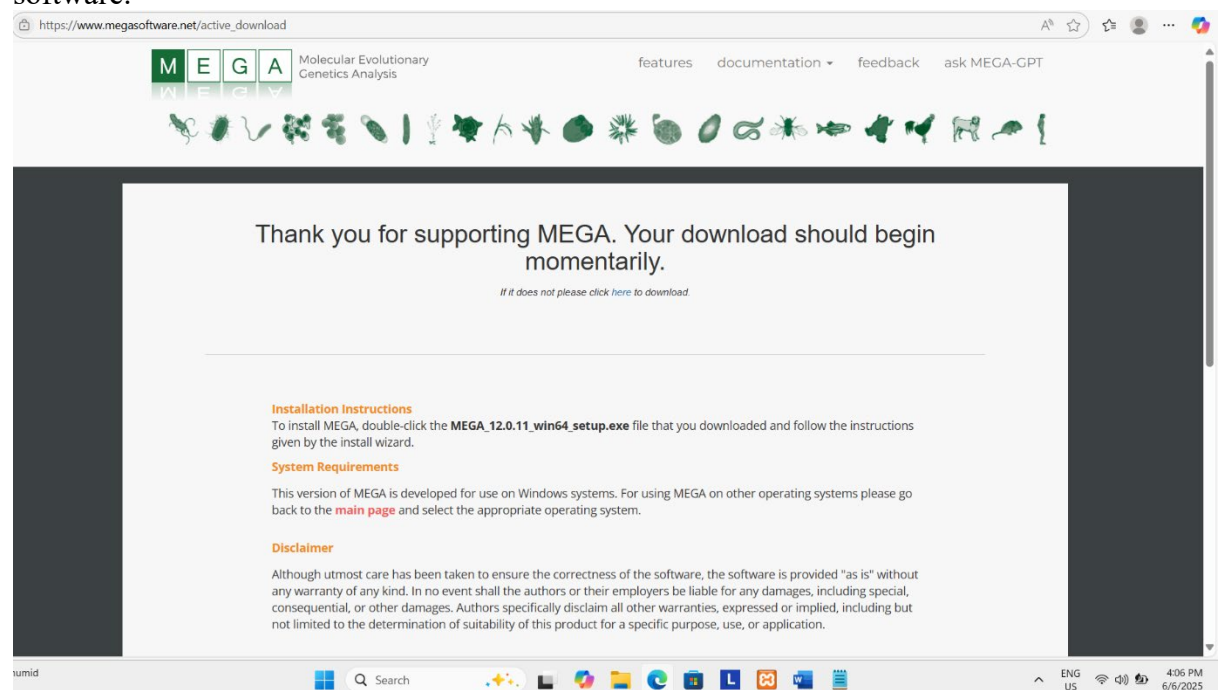
Name : Koo Xuan (A23CS0300)

1) Determine whether human and chimpanzee mitochondrial DNA sequences have equal evolutionary rates between lineages. To do this, use Tajima's relative rate test as implemented in MEGA

i) Obtain MEGA software Perform pairwise alignment at the NCBI BLAST website.

Solutions/comments:

MEGA software is available from <http://www.megasoftware.net>. Download MEGA software:



ii) Obtain mitochondrial DNA sequences from human, chimpanzee, bonobo, orangutan, gorilla, and gibbon

Solutions/comments:

Go to NCBI Nucleotide Database and download FASTA format file of full sequence of mitochondrial DNA sequences from human, chimpanzee, bonobo, orangutan, gorilla, and gibbon.

Human:

The screenshot shows the NCBI Nucleotide Database interface. The search results for "Homo sapiens mitochondrion, complete genome" are displayed. The FASTA format is selected for download. The download menu is open, showing options for "Complete Record", "Coding Sequences", and "Gene Features". The "Choose Destination" section shows "File" selected. The "Download 1 item." section shows "Format" set to "FASTA". The "Show GI" checkbox is unchecked. The "Create File" button is visible.

Chimpanzee:

The screenshot shows the NCBI Nucleotide Database interface. The search results for "Pan troglodytes mitochondrion, complete genome" are displayed. The FASTA format is selected for download. The download menu is open, showing options for "Complete Record", "Coding Sequences", and "Gene Features". The "Choose Destination" section shows "File" selected. The "Download 1 item." section shows "Format" set to "FASTA". The "Show GI" checkbox is unchecked. The "Create File" button is visible.

Bonobo:

Orangutan:

Gorilla:

https://www.ncbi.nlm.nih.gov/nucleotide/X93347.1?report=fasta

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Log in

Nucleotide

Advanced

FASTA

Gorilla gorilla gorilla mitochondrial DNA, complete genome
GenBank: X93347.1
[GenBank](#) [Graphics](#)

>X93347.1 Gorilla gorilla gorilla mitochondrial DNA, complete genome
GTTTATGTAGCTTAACCGCTAAAGCAAACTGAAATGTAAGATGGGCTGCCCGCCCATAAACA
AATAGGTTTGGCTCTAGCTTCTTATTAACCTTTAGTAGGATTACACATGCAAGATCCCGCCCAAGT
GAGTCACCTCTTAATACACGATCAAAAGGAACAAGCATCAAGTACGAGAAATGCAAGTCAAAACG
TTAGCTAGCACACCCCGGAGAGACAGTGTAAACCTTTAGCAATAAACGAAGTTTAACTAAG
CAATCACTAACCCCAAGGTTGGTCAATTTGGTGCAGCCACCGGCTCACAGATTAAACCAAGCAATAG
AAATCGGCTAAAGAGTTTATAGATCAATCCCCCAATAAAGCTAAATTCACCTGAGTTGTAAAAACT
CCAGCTGATATAAATAAATACACGAAGTGGCTTTAATATATCTGAACACACAATAGCTAGGACCCAAC
TGGAGTAGATACCCACTATGCTAGCTTAACCTTGAAGCTTGAAGTAAATTAACAGAGCTGCTGCCGAA
CACTACGAGCAGAGCTTAAACCTCAAGGACCTGGGCGTCTTACATCTCTTAGAGGAGCTGTTCT
GTAATCGATAAACCCGATCAACCTCACACCTCTTGGTCAGCCTATATACCGCATCTTCAGCAACCC
TGACGAAGGCACAAAGTAGACAGATGACCTACGTAAGAGTGTAGTCAAGGCTAGCTCCATGAGGTG
GCAAGAAATGGGCTACATTTCTACTTCGAAACCTACGATAACCTTATGAACCTAAGGAGTGAAGGT
GGATTTAGCACTAACTAAGAGTAGAGTGTAGTTGAACAGGGCCCTGAAGCGCTACACACGCCCGT
CACCTCTCTCAAGTATACTTCAAGGACATTTAACTAAACCCCTACGCATCTATATAGAGGAGATAGT
CGTAACTGGTAAAGTGTACGGAAGTGCATCTGGACGAACAGAGTGTAGCTTAAACAAAGACCCAA
CTTACCTTAGAGGATTTCACTCACTTGAACCTCTGAGCAAACTTAGCCCCAAACCACTCCACAT
TACTACCAACCACTTTAATCAACCATTTACCAAAATGAAGTAGGCGATAGAGATGTTAATATGGG
CAATAGATATAGTACGCAAGGAGAGTGAAGAAATAACCAAGCAGACAGCAGAGGACAAACCC
TGTACCTCTGCATAATGAATTAAGTGAATAATTTGCAAGAGAACCGAGCTAAGACCCCGCAAC
CAGACGAGCTACCTAAGACAGCTAAGAGACACCCGCTATGTAGCAAAATAGTGGGAAGTTTCCTA
GGTAGAGGCACAAACCTACGAGCCTGGTGTAGCTGGTGTCTCAAGACAGAACTTAGTTCACTTTA
AATTTACCCAGAGACCTCCCAATCTCTCGTAAATTTAACTGTGTAGTCAAGAGGAGACAGCTCTTGG
GACACTAGGAAAAAACCCTGTAAAGTAGTAAAAAATTTAATACCATAGTAGGCTTAAAGCAGCCACC

Send to:

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☐ Gene Features

Choose Destination
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☐ Clipboard
☐ Analysis Tool

Download 1 item.
Format:
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the ND4L gene
mitochondrial diversity within
due to extinction (Sci Rep. 2018)

More about the gene ND4L
ND4L gene
Also Known As: KEG83_p05

Related information
BioProject

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Gibbon:

https://www.ncbi.nlm.nih.gov/nucleotide/AB504751.1?report=fasta

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Nucleotide

Advanced

FASTA

Nomascus siki mitochondrial DNA, complete genome
GenBank: AB504751.1
[GenBank](#) [Graphics](#)

>AB504751.1 Nomascus siki mitochondrial DNA, complete genome
GTTTATGTAGCTTAACCGCTAAAGCAAACTGAAATGTAAGATGGGCTGCCCGCCCATAAACA
AATAGGTTTGGCTCTAGCTTCTTATTAACCTTTGTAAGATTACACATGCAAACTCCCGCCCAAGT
AAATGGCTCTCAAACTCGTCACGATCAAGAGGAGGCGGTATCAAGCAGCAGCAATGCAAGCTCAAAACA
CCCAGCTAGCCACACCCCGGAGAAACAGCAGTGTAAACCTTTAGCAATAAACGAAGTTTAACTAAG
GCTATACCACTCAGGGCGGTCACTTCTGCGAGCACCGCGCTATACGATGCTTACCTAGTCAATAG
AACCGGCTAAAGAGTGTTTAGATCATCTCCAAATAAGCTAAGCTCCATCAAGTTGTAAAAACTC
TGGCTGGCTAAATTAACCTAGGAAGTGGCTTTAAACCTCTGAATACCAATAGCTAAGACCAACT
GGGATTAGATACCCCACTGCTAGCTTAACTTCAACAGCTAGATCAACAGACTGCTGCCGAAC
ACTACGAGCAACAGCTTAAACTCAAGGACCTGGGCGTGTCTCACACCCCTAGAGGAGCTGTCTTA
TAATCGATAAACCCCGTCAACCTCACATCTTGGTCAGCCTATATACCGCATCTTCAGCAAACTCT
GCAAGAGGCTATAAGTAAAGCGCAACACCCACATAAAGACCTTAGTCAAGGTGTAGCTTAACTGAGGATGG
GAGAGATGGCTACATTTTCACTCTGAGAAACCTAGATTAACCTTATGAATTTCAAGTGTGAGGAG
GGATTAGAGTAAATTAAGATAGAGTGTAAATTAAGCAAGGCTTGAAGCGCTACACACCGCCGT
CACCTCTCTCAAGTATATTCAAGGAACCCCACTAAGCCCGCGCATCTATATAGAGGAGATAAG
TGTAACTGTAGTGTAGCTGGAAGTGGCTTGGATTAACCAAGGCTAGCTTAAACAAAGCAGCCCG
GCTTACCTGGGAGATTCACTAGACTTGAACCTCTGAGCAACCTAGCCCAAGTTTCCCGATC
CTACTATTAAACAGCATCAACCAACATTTATACACATTAAGTATAGGCGATAGAAATTTAATATGG
CGCATAGATAGTACGTCGCAAGGAGAAAGCAAAACCAACCAAGCAGCAAAACAGCAGGATAGCTCC
TGTACTCTTGCATATGAATTAACCAACCACTTACGAGGAGACCAAGCTAAGACCCCGCAAC
CAGACGAGCTACCTAAGAACCGTGAAGAGCACACCGCTGTATGAGCAAAATAGTGGGAAGATTATA
GGTAGAGGTGACAAAGCTCAGGCGCTGGGATAGTGGTTCGAAGATAGAACTTGTTCACATTTA
AATTTACCCAGAGACCCCAATCTCTCGTAAATTTAACTGTGTAGTCAAGAGGAGACAGCTCTTAC
ACACTAGGAAAAAACCCTGCAAGAGAGTAAAAACAAACCCATAGTTGGCTTAAAGCAGCCACCA

Send to:

☒ Complete Record
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☐ Gene Features

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☐ Clipboard
☐ Analysis Tool

Download 1 item.
Format:
Show GI ☐

Full text in PMC
Gene
PubMed (Weighted)

Recent activity
Turn Off Clear

Nomascus siki mitochondrial DNA, complete genome

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iii) Apply Tajima's test using an appropriate outgroup. Is the probability value significant (<0.05)?

Solutions/comments:

After obtained the FASTA sequence, use **NCBI BLAST** for alignment. Choose "Align two or more sequences" and Human sequence as "Query" and Chimpanzee as "Subject". Then submit and scroll to see alignment and similarity.

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Important update
Effective August 2025, the **ClusteredNR** database will become the default Protein BLAST database. [Learn more about ClusteredNR](#)

blastn blastp blastx tblastn tblastx

Align Sequences Nucleotide BLAST

BLAST® programs search nucleotide subjects using a nucleotide query. [more...](#)

[Reset page](#) [Bookmark](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

Query subrange [?](#)

From

To

Or, upload file [Choose File](#) human sequence fasta [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☒ Align two or more sequences [?](#)

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

Subject subrange [?](#)

From

To

Or, upload file [Choose File](#) chimpanzee sequence fasta [?](#)

Program Selection

Optimize for

☒ Highly similar sequences (megablast)

☐ More dissimilar sequences (discontiguous megablast)

☐ Somewhat similar sequences (blastn)

Choose a BLAST algorithm [?](#)

BLAST Search nucleotide sequence using Megablast (Optimize for highly similar sequences)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with + sign

[+ Algorithm parameters](#)

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Search

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US

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Here is the results:

https://blast.ncbi.nlm.nih.gov/Blast.cgi

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BLAST® » blastn suite-2sequences » results for RID-459796JY114

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Job Title **NC_012920.1 Homo sapiens mitochondrion, complete...**

RID [459796JY114](#) Search expires on 06-07 16:20 pm [Download All](#)

Program Blast 2 sequences [Citation](#)

Query ID IclQuery_7617935 (dna)

Query Descr NC_012920.1 Homo sapiens mitochondrion, complete genome

Query Length 16569

Subject ID IclQuery_7617937 (dna)

Subject Descr NC_001643.1 Pan troglodytes mitochondrion, complete g...

Subject Length 16554

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Filter Results

Percent Identity to E value to Query Coverage to

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Descriptions [Graphic Summary](#) [Alignments](#) [Dot Plot](#)

Sequences producing significant alignments [Download](#) [Select columns](#) [Show](#) [?](#)

☒ select all 1 sequences selected [Graphics](#) [MSA Viewer](#)

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ NC_001643.1 Pan troglodytes mitochondrion, complete genome		21837	22442	100%	0.0	91.33%	16554	Query_7617937

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Next, we compare with Bonobo and Orangutan.

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Align Sequences Nucleotide BLAST

blastnblastpblastxtblastntblastx

BLASTN programs search nucleotide subjects using a nucleotide query. more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

Query subrange [?](#)

From

To

Or, upload file

Choose File

bonobo sequence.fasta [?](#)

Job Title

NC_012920.1 Homo sapiens mitochondrion, complete...

Enter a descriptive title for your BLAST search [?](#)

☒ Align two or more sequences [?](#)

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

Subject subrange [?](#)

From

To

Or, upload file

Choose File

orangutan sequence.fasta [?](#)

Program Selection

Optimize for

☒ Highly similar sequences (megablast)
 ☐ More dissimilar sequences (discontiguous megablast)
 ☐ Somewhat similar sequences (blastn)

Choose a BLAST algorithm [?](#)

BLAST

Search nucleotide sequence using Megablast (Optimize for highly similar sequences)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with a sign

+ Algorithm parameters

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Search

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Search Summary [v](#)

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Job Title

NC_012920.1 Homo sapiens mitochondrion, complete...

RID

459ZYS1U114

Search expires on 06-07 16:34 pm

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Program

Blast 2 sequences

Citation [v](#)

Query ID

lcl|Query_81953 (dna)

Query Descr

NC_001644.1 Pan paniscus mitochondrion, complete genome

Query Length

16563

Subject ID

lcl|Query_81955 (dna)

Subject Descr

X97708.1 Pongo abelii (YN91-227) mitochondrial DNA for ...

Subject Length

1016

Other reports

MSA viewer [?](#)

Filter Results

Percent Identity

to

E value

to

Query Coverage

to

Filter

Reset

Descriptions

Graphic Summary

Alignments

Dot Plot

Sequences producing significant alignments

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☒ select all

1 sequences selected

Graphics

MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per Ident	Acc. Len	Accession
☒ X97708.1 Pongo abelii (YN91-227) mitochondrial DNA for control region		508	508	4%	1e-146	79.86%	1016	Query_81955

sunrny

Search

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Last, we compare with Gorilla and Gibbon.

6

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Align Sequences Nucleotide BLAST

blastnblastpblastxtblastntblastx

BLASTN programs search nucleotide subjects using a nucleotide query. more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

Query subrange

From

To

Or, upload file

Choose File

gorilla sequence.fasta

Job Title

NC_012920.1 Homo sapiens mitochondrion, complete...

Enter a descriptive title for your BLAST search

☒ Align two or more sequences

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

Subject subrange

From

To

Or, upload file

Choose File

gibbon sequence.fasta

Program Selection

Optimize for

☒ Highly similar sequences (megablast)
☐ More dissimilar sequences (discontiguous megablast)
☐ Somewhat similar sequences (blastn)

Choose a BLAST algorithm

BLAST

Search nucleotide sequence using Megablast (Optimize for highly similar sequences)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with a sign

Algorithm parameters

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Search

ENG US

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6/6/2025

https://blast.ncbi.nlm.nih.gov/Blast.cgi

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Search Summary

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Job Title

NC_012920.1 Homo sapiens mitochondrion, complete...

RID

45A2HF93114

Search expires on 06-07 16:35 pm

Download All

Program

Blast 2 sequences

Citation

Query ID

lclQuery_168727 (dna)

Query Descr

X93347.1 Gorilla gorilla gorilla mitochondrial DNA, comple...

Query Length

16412

Subject ID

lclQuery_168729 (dna)

Subject Descr

AB504751.1 Nomascus siki mitochondrial DNA, complete ...

Subject Length

16476

Other reports

MSA viewer

Filter Results

Percent Identity

to

E value

to

Query Coverage

to

FilterReset

Descriptions

Graphic Summary

Alignments

Dot Plot

Sequences producing significant alignments

Download

Select columns

Show 100

select all

1 sequences selected

Graphics

MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
AB504751.1 Nomascus siki mitochondrial DNA, complete genome		14879	15349	97%	0.0	84.01%	16476	Query_168729

ps://blast.ncbi.nlm.nih.gov/Blast.cgi#

ly sunny

Search

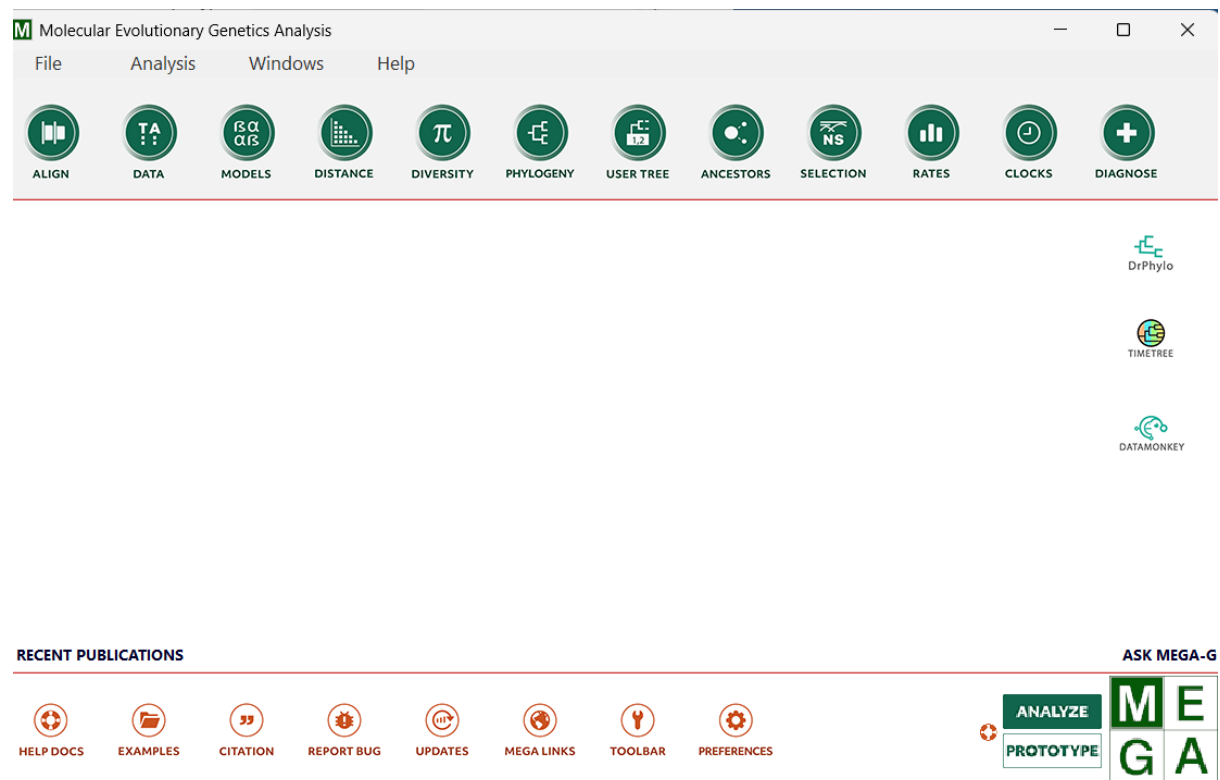
ENG US

4:35 PM

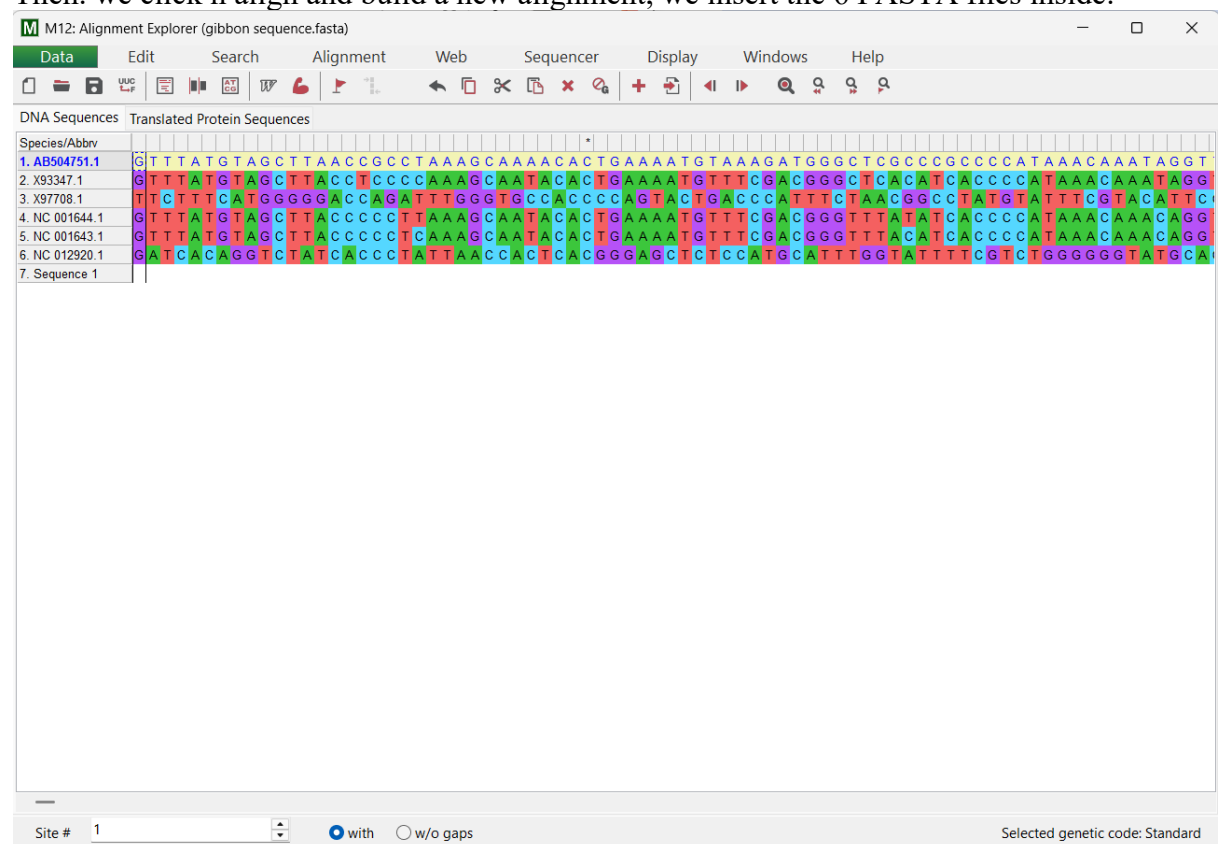
6/6/2025

So now to apply Tajima's test using an appropriate outgroup. First we open MEGA app.

7



Then. we click n align and build a new alignment, we insert the 6 FASTA files inside:



To continue, we test human, chimpanzee and outgroup as bonobo first .

M12: Analysis Preferences

Tajima's Relative Rate Test

Option	Setting
ANALYSIS	
Scope	For 3 chosen sequences
CHOOSE TAXA TRIPLET	
Taxon A →	NC 012920.1
Taxon B →	NC 001643.1
Outgroup →	NC 001644.1
SUBSTITUTION MODEL	
Substitutions Type →	Nucleotide
Substitutions to Include →	All
DATA SUBSET TO USE	
Gaps/Missing Data →	Complete deletion
Select Codon Positions →	☒ 1st ☒ 2nd ☒ 3rd ☒ Noncoding Sites

And here is the result for human, chimpanzee and bonobo

M Results

File Edit View Windows Help

TXT

Table. Results from the Tajima's clock test for 3 Sequences.

Configuration	Count
Identical sites in all three sequences	1406
Divergent sites in all three sequences	5443
Unique differences in Sequence A	3376
Unique differences in Sequence B	3107
Unique differences in Sequence C	3221

NOTE.-- The comparative analysis investigated the equality of evolutionary rates between sequences **A** (NC 012920.1) and **B** (NC 001643.1), utilizing sequence **C** (NC 001644.1) as an outgroup in Tajima's relative rate test [1]. The χ^2 test yielded a value of 11.16 ($P = 0.00084$ with 1 degree[s] of freedom). Conventionally, a P -value less than 0.05 is employed to test the null hypothesis of equal rates between the selected lineages. The analytical procedure encompassed 3 coding nucleotide sequences using 1st, 2nd, 3rd, and non-coding positions. The complete deletion option was applied to eliminate positions containing gaps and missing data resulting in a final data set comprising 16,553 positions. Evolutionary analyses were conducted in MEGA12 [2].

1. Tajima F. (1993). Simple methods for testing molecular clock hypothesis. *Genetics* **135**:599-607.
2. Kumar S., Stecher G., Suleski M., Sanderford M., Sharma S., and Tamura K. (2024). Molecular Evolutionary Genetics Analysis Version 12 for adaptive and green computing. *Molecular Biology and Evolution* **41**:1-9.

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Then, we test Orangutan, Gorilla and outgroup as Gibbon.

M12: Analysis Preferences

Tajima's Relative Rate Test

Option	Setting
ANALYSIS	
Scope	For 3 chosen sequences
CHOOSE TAXA TRIPLET	
Taxon A →	X97708.1
Taxon B →	X93347.1
Outgroup →	AB504751.1
SUBSTITUTION MODEL	
Substitutions Type →	Nucleotide
Substitutions to Include →	All
DATA SUBSET TO USE	
Gaps/Missing Data →	Complete deletion
Select Codon Positions →	☒ 1st ☒ 2nd ☒ 3rd ☒ Noncoding Sites

Here is the result:

M Results

File Edit View Windows Help

TXT

Table. Results from the Tajima's clock test for 3 Sequences.

Configuration	Count
Identical sites in all three sequences	113
Divergent sites in all three sequences	283
Unique differences in Sequence A	333
Unique differences in Sequence B	143
Unique differences in Sequence C	144

NOTE.-- The comparative analysis investigated the equality of evolutionary rates between sequences **A** (X97708.1) and **B** (X93347.1), utilizing sequence **C** (AB504751.1) as an outgroup in Tajima's relative rate test [1]. The χ^2 test yielded a value of 75.84 ($P = 0.00000$ with 1 degree[s] of freedom). Conventionally, a P -value less than 0.05 is employed to test the null hypothesis of equal rates between the selected lineages. The analytical procedure encompassed 3 coding nucleotide sequences using 1st, 2nd, 3rd, and non-coding positions. The complete deletion option was applied to eliminate positions containing gaps and missing data resulting in a final data set comprising 1,016 positions. Evolutionary analyses were conducted in MEGA12 [2].

1. Tajima F. (1993). Simple methods for testing molecular clock hypothesis. *Genetics* **135**:599-607.

2. Kumar S., Stecher G., Suleski M., Sanderford M., Sharma S., and Tamura K. (2024). Molecular Evolutionary Genetics Analysis Version 12 for adaptive and green computing. *Molecular Biology and Evolution* **41**:1-9.

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2) Perform phylogenetic analyses using MEGA software

- i) Go to the conserved domain Database (<http://www.ncbi.nlm.nih.gov/cdd>) at NCBI. Click the box to set the view to default tracks.
- ii) Enter lipocalins (or another family of your choice; you can also begin at Ensembl, HomoloGene, or Pfam).
- iii) Select the mFasta format then click “Reformat.” The result is a multiple sequence alignment. Copy this into a text editor (such as NotePad++), and simplify the names of the sequences.
- iv) Import the file (or paste the sequences) into MEGA as shown in Fig. 7.9. Align the sequences, save in the .mas and .meg formats.
- v) Choose Phylogeny > Construct/Test to create neighbor-joining, maximum likelihood, or other trees.
- vi) For each tree you create, read the caption. Try the tree tools (e.g. placing a root, flipping nodes, showing or hiding branch lengths, interconverting display formats).
- vii) Perform bootstrapping. Identify clades having low levels of support. Why does this occur?

Solutions/comments:

From the home page of NCBI enter lipocalin; follow the link of the search results to conserved domains; view the result.

The screenshot shows the NCBI Conserved Domains database search results for the query 'lipocalin'. The search was performed using the 'Conserved Domains' search tool. The results page displays a table of resources, including the Conserved Domain Database (CDD), CD-Search, and SPARCLE. The CDD entry is highlighted, showing a list of conserved domains and their associated protein sequences. The CD-Search entry is also visible, providing information about the search tool and its results. The SPARCLE entry is also present, detailing the Subfamily Protein Architecture Labeling Engine (SPARCLE) and its use in protein classification. The page also includes a 'Highlights' section with a 3-D structure of a conserved domain and a list of conserved features (binding and catalytic sites). The search results are displayed in a table format, with columns for the resource name, description, and a link to the full record. The table includes entries for CDD, CD-Search, and SPARCLE. The CDD entry is the most detailed, showing a list of conserved domains and their associated protein sequences. The CD-Search entry provides information about the search tool and its results. The SPARCLE entry details the Subfamily Protein Architecture Labeling Engine (SPARCLE) and its use in protein classification. The page also includes a 'Highlights' section with a 3-D structure of a conserved domain and a list of conserved features (binding and catalytic sites). The search results are displayed in a table format, with columns for the resource name, description, and a link to the full record.


```

mgsshhhhhssglvprgshmasmtggqqmgrgsdniqvqenfnisriYGKWYNLAIGStcpwlKKIM
DRMTVSTLVLGE
GaTEAEISMTSTRWRKGVCEETSGAYEKTDTDGKFLYHKSKNITM---
ESYVVHTNYDEYAIFLTKKFSRhHgPTITAK
LYGRAPQLRETLlq-dFRVVAQGVGIPEDSIFTMADRGeCvpgeq-----
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-----
>gi|81886784|sp|O70160|O70160_CAVPO
-----
mwgaealllllgsclaisvnpvltlppddiqvqenfdesrmYGKWYSLAIGStcpwlKRIKDRLSVDTM
VLGK
GaTEAQMSTTRTHWRRGVCERTSGFYEKTDTAGKYLYSNPKWNLTM---
ESYVVHTNYDEYATVLTKKFSRhHgPTITAK
LYGREPQLRDSLlq-
gFREMALS VGIPEDSIFTMANRGecipgeqipqptpalrarravllqedegsgagqpvt dftkke
dscqlghsagpclglfkryfyngssmaceifhyggclngnnfnsekeclqtcrtvaacnlpivpgpc
qgsaqlwaf dai
kgkcvrftyggcgngnkfysekeckeycgapgdgdeellrfsn
>gi|82123701|sp|O93588|O93588_LACVV
-----
lvfgmtpdyifpvsadipvvpnf dpqktVGKWHPIGMASKlpelTPYEQKISPMDHIVEV
--IDGDMKLTANYMSDGVSK EATAMLKHTDKPGVFKFT-----DG---
EVHVLDVDFEKYIMLYVKKS-----SHEALF
LSARGPDVEDDIke-kFKKLVLEQSFPEANIKYFNAEQctptaa-----
-----
-----
-----
-----
>gi|82071154|sp|O93589|O93589_LACVV
-----
mtpdyifpvsadipvvanf dtpktVGKWHPIGMASKlpelTPYEQKISPMDHIVEV
--TDGDMKLTANYMSDGVCKTSVLVLKHTDKPGVFKVP-----DG---
EVHVIKMLILKSISFFTSK----KpTHEALF
LSARGSTGGDDIka-kFKKLVWEQIILEAHIKYLNVEQctpkag-----
-----
-----
-----
-----
>gi|190359792|sp|A2AJB7.1|LCN5_MOUSE
-----
mcsvarhmesimlftllglcv glaagteaavvkdfdv nkflGFWYEIALASKmgayGLAHKEEKMGAM
VVEl
--KENLLALTTTYNEGHCVLEKVAATQVDGSAKYKVTR--ISGEK---
EVVVVATDYMTYTVIDITSLVA-GaVHRAMK
LYSRSLDNNGEAln-nFQKIALKHGFSETDIHILKHDLtcvnalqsgqi-----
-----
-----
-----
-----
>gi|62286893|sp|Q924P3.1|LCN8_MOUSE
-----
mearllsnvcgfflvfllq aestrvelvpekiAGFWKEVAVASdqklvLKAQRRVEGLFLTF--
--SGGNVTVKAVYNSSGSCVTESSLGSE RDTVGEFA-----FPGNR---
EIHVLDTDYERYTILKLTLLWQ-GrNFHVLK
YFTRSLENEDEPgfwlFREMTADQGL-----YMLARHgrcaellkeglv-----
-----

```

```
>gi|149242697|pdb|2OVA|A
```

```
qkplrprpaspistiqpkanfdaqfAGTWLLVAVGSaarflQEQGHRAEATTLHVAP
--QGTAMAVSTFRKLDGICWQVRQLWGDGTGVLGRFLLQ---
ARGARgavHVVVAETDYQSFVLYLERA-----GQLSVK
LYARSLPVSDSVls-gFEQVRVQEAHLTEDQIFYFPKYGfceaadqfhvldevrr-----
```

```
>gi|365813052|pdb|3SAO|B
```

```
tvprdrsevAGKWIIVALASntdfLAEKGKMKVMARISF
L-GEDELEVSYAAPSPKGCWKWETTFKKTSDDGEVYYS---EEAEK---
TVEVLDTDYKSYAVIFATRV-KdGrTLHMMR
LYSRSEVSPtAma-iFRKLARERNYTDEMVAVLPSQaacsdevlvpr-----
```

```
>gi|82216128|sp|Q8UWH4|Q8UWH4_CHICK
```

```
mqplpqhdpdkarmtavlpvlvtllccllrvgaevpvqldfdtkkfAGMWHMMAAVSncpvfLGMKDK
MTSSATAVSF
T-PEGHMTMETVIPLPEECKKIMMLFQQSGQAGHYIGTE--KGEKR---
DLRVMDTDYEHYAIVYSQQKDS-QePSTMLQ
LYTRNQDMSPQLlq-kFKELFHSMGLTEDMLVVLPHSDrcaktqk-----
```

Adjust the file names to make them more informative. You can look up a string of identifiers using BioMart, or biomaRt, or EDirect, or by entering a string at once into an NCBI web search.

```
>human_tear_lipocalin
```

```
-----asdeeiqdvSGTWYLKAMTVdrefpEMNLE--
SVTPMTLTTL-EGGNLE
AKVTMLISGR--CQEVKAVLEKTDEPGKYTA-----DGGKHVAYIIRSHVKDHYIFYSEGELHG-
K-PVRGVKLVGRD
PKNNLEAledFEKAAGARGLSTESILIPRQSEtcspgsawshpqfek-----
```

```
>mouse_allergen
```

```
-----mqfQGEWFLGLADnt-
frREHRAl1NFFTTLFELK-EKSQFQ
VTNSMTRGKH--CNTWSYTLIPATKPGQFTRdnrgsgpGADRENIQVIETDYITFALVLSLRQTSs-
Q-NITRVSLLRN
WRLSHKtidkfICLTRTQNLTKDNFLFPDLSDwlpdpqvc-----
```

```

>mouse_siderocalin
-----qdstqnlipapslltvplqpdfrsdqfRGRWYVVGLAGNa-
vqKKTEGsftMYSTIYELQ-ENNSYN
VTSILVRDQDqgCRYWIRTFVPSSRAGQFTLgnmhrypQVQSYNVQVATTDYNQFAMVFFRKTSENkQ
-YFKI-TLYGRT
KELSPeLkerFTRFAKSLGLKDDNIIFSVPtDqcidnsawshpqfek-----
-----
-----
-----
-----
>human_siderocalin
-----qdstsdlipapplskvplqqnfqdnqfQGKWYVVGLAGNa-
ilREDKDpqKMYATIYELK-EDKSYN
VTSVLFRKKK--
CDYWIRTFVPGSQPGEFTLgniksypGLTSYLVRVVSTNYNQHAMVFFKKVSQNrE-YFKI-TLYGRT
KELTSELkenFIRFSKSLGLPENHIVFPVPIDqcidg-----
-----
-----
-----
-----
> lizard_Androgen-regulated epididymal secretory protein
-----
lvfgmtpdyifpvsadipvvpnfdpcktVGKWHPIGMASklpelTPYEQkiSPMDHIVEV--IDGDMK
LTANYMSDGV--SKEATAMLKHTDKPGVFKFt-----DGEVHVLDVDFEKYIMLYVKKS-----
-SHEALFLSARG
PDVEDDIkekFKKLVLVEQSFPEANIKYFNAEQctptaa-----
-----
-----
-----
-----
> lizard2_Androgen-regulated epididymal secretory protein
-----
mtpdyifpvsadipvvanfdtpktVGKWHPIGMASklpelTPYEQkiSPMDHIVEV--TDGDMK
LTANYMSDGV--CKTSVLVLKHTDKPGVFKVp-----DGEVHVILKMLILKSISFFTSKk----
P-THEALFLSARG
STGGDDIkakFKKLVLWEQIILEAHIKYLNVEQctpkag-----
-----
-----
-----
-----
>mouse_Epididymal retinoic acid-binding protein
mcsvarhmesimlftllglcvglaagteaavvkdfdvknfLGFWYEIALASkmgayGLAHKeeKMGAM
VVEl--KENLLA
LTTTYYNegH--CVLEKVAATQVDGSAKYKVt-----rISGEKEVVVVATDYMTYTVIDITSLVAg-
A-VHRAMKLYSRS
LDNNGEAlnnFQKIALKHGFSETDIHILKHDLtcvnaIqsgqi-----
-----
-----
-----
-----
>bovine_Alpha-1-microglobulin
mrsIsgllllltaclavnassvptlpddiqvqenfdlsriYGKWFNVAVGStcpwlKRFKEkmTMSTV
VLIAGpTSKEIS
VTNTHRRKGV--CESISGTYEKTSADGKFLYhk---
akWNITMESYVVHTNYDEYAIFLTkKLSRRhG-PTITVKLYGRE
PQLRESLleeFREVALGVGIPEDAIFTMPDRGecvpgeqdpvptplsrarravltqeeegsgagqpvt
nfskkadscqld
ysqgpcIglfIkryfyngtSmacetflyggcmgnngnflsekeclqtctrtveacnlpivqgpcrsyiqI

```


The alignment explorer of MEGA shows the imported multiple alignment from NCBI's Conserved Domain Database. Here note the asterisks mark 100% conservation of amino acid sequence in a given column; the GXW motif is evident.

M12: Alignment Explorer (human_Alpha-1-microglobulin.fasta)

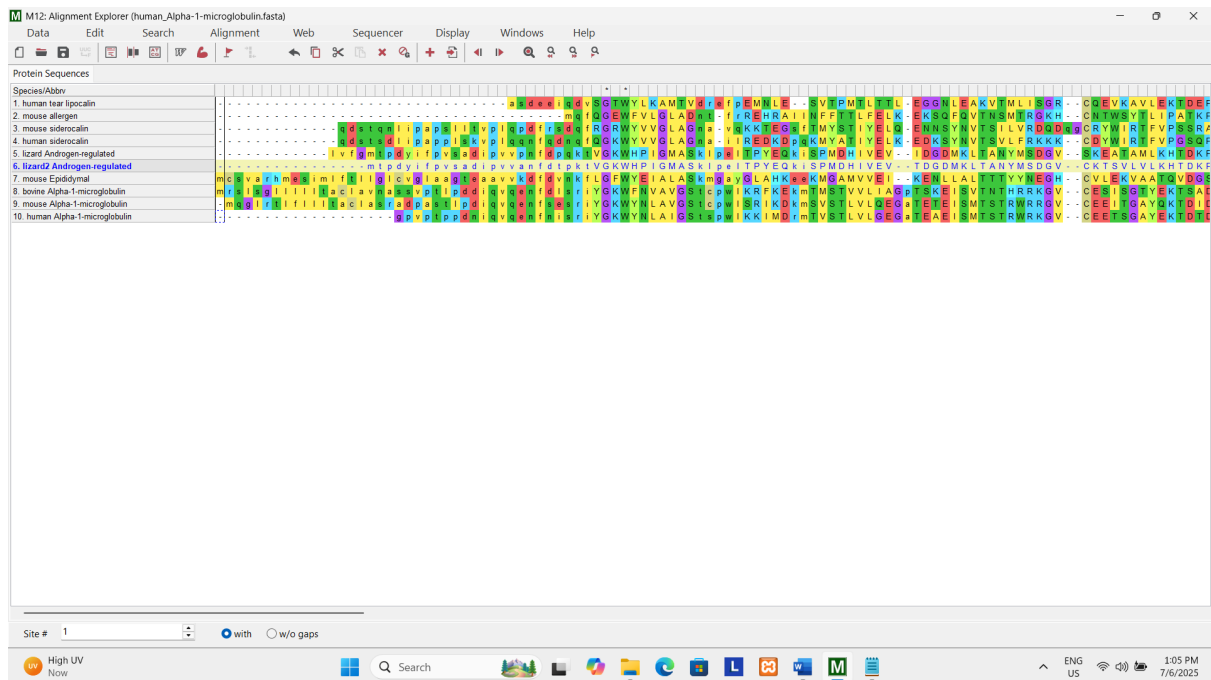
Species/Abbrev:

1. human tear lipocalin
2. mouse allergen
3. mouse siderocalin
4. human siderocalin
5. lizard Androgen-regulated
6. lizard2 Androgen-regulated
7. mouse Epididymal
8. bovine Alpha-1-microglobulin
9. mouse Alpha-1-microglobulin
10. human Alpha-1-microglobulin

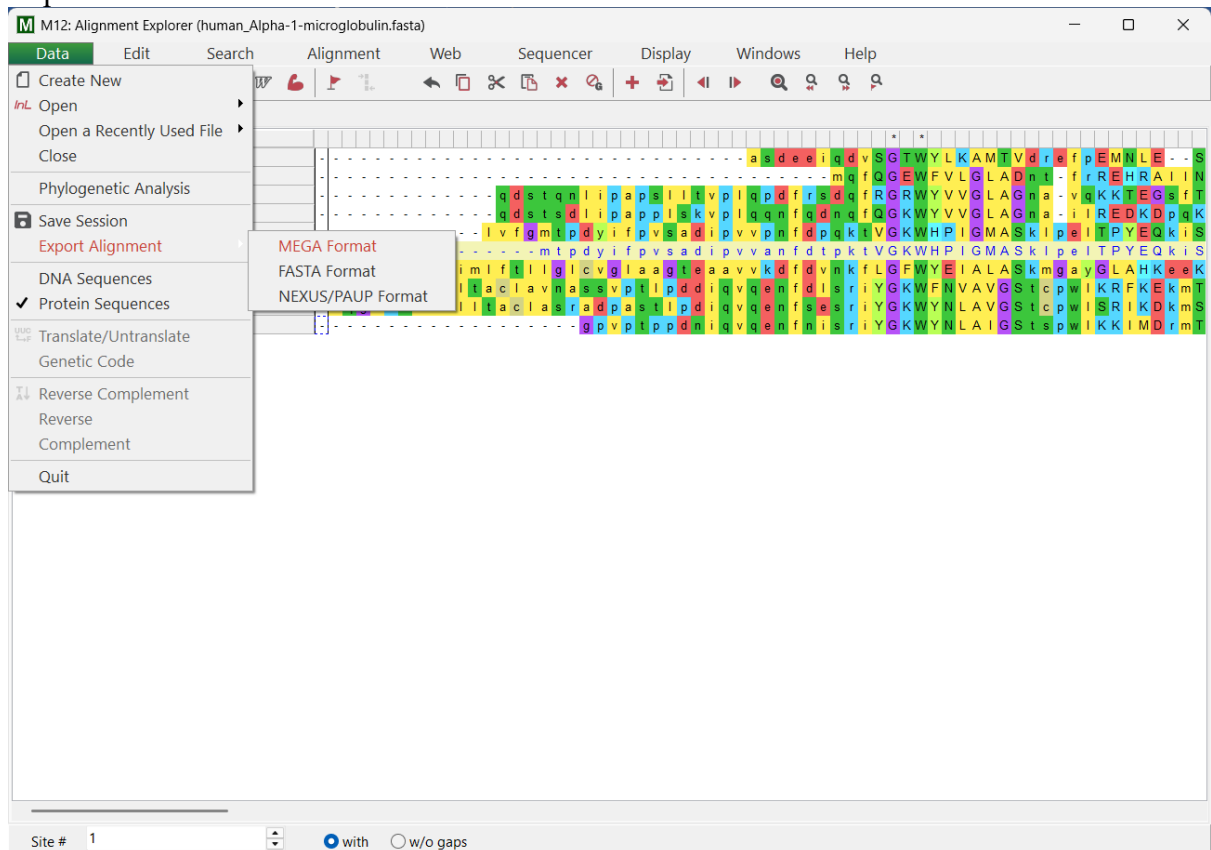
Site # 1

with w/o gaps

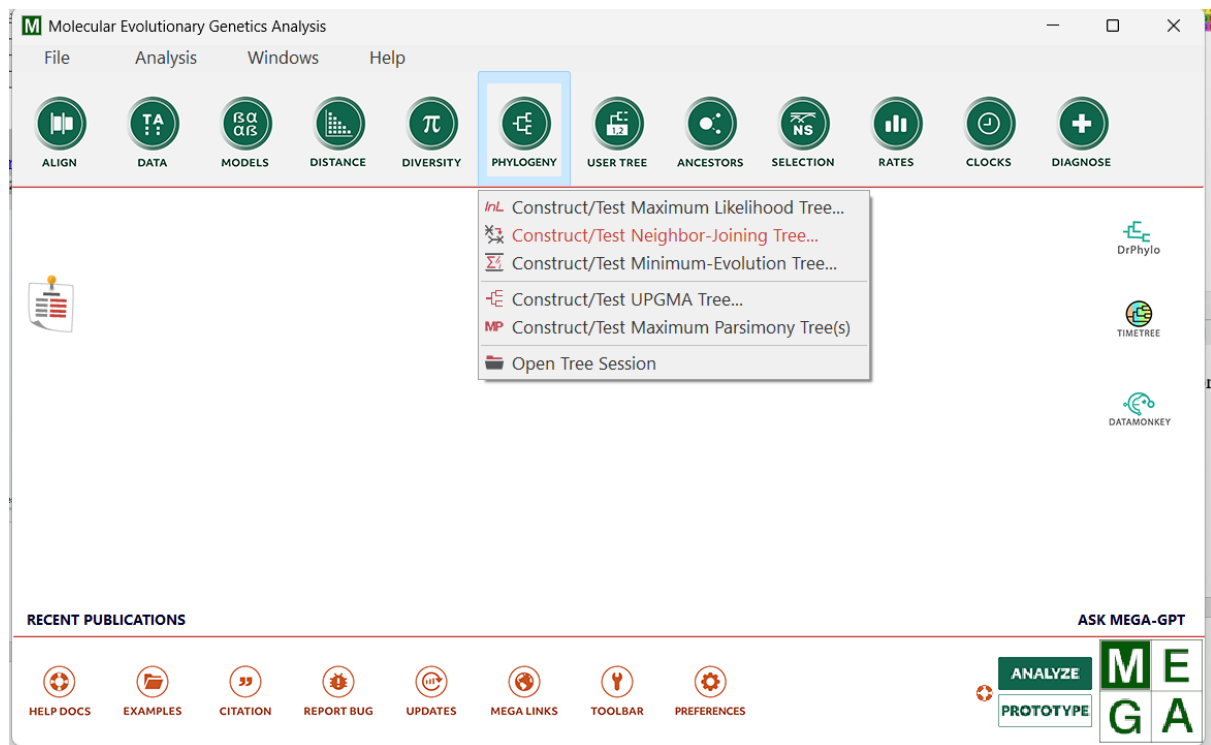
ENG US 1:05 PM 7/6/2025



Export the file to MEGA file:



Choose Phylogeny (top menu) and select the method neighbor-joining of tree construction :



The analysis preferences dialog allows you to select a variety of parameters of the analysis, such as bootstrapping (under “phylogeny test”).

M12: Analysis Preferences

Phylogeny Reconstruction

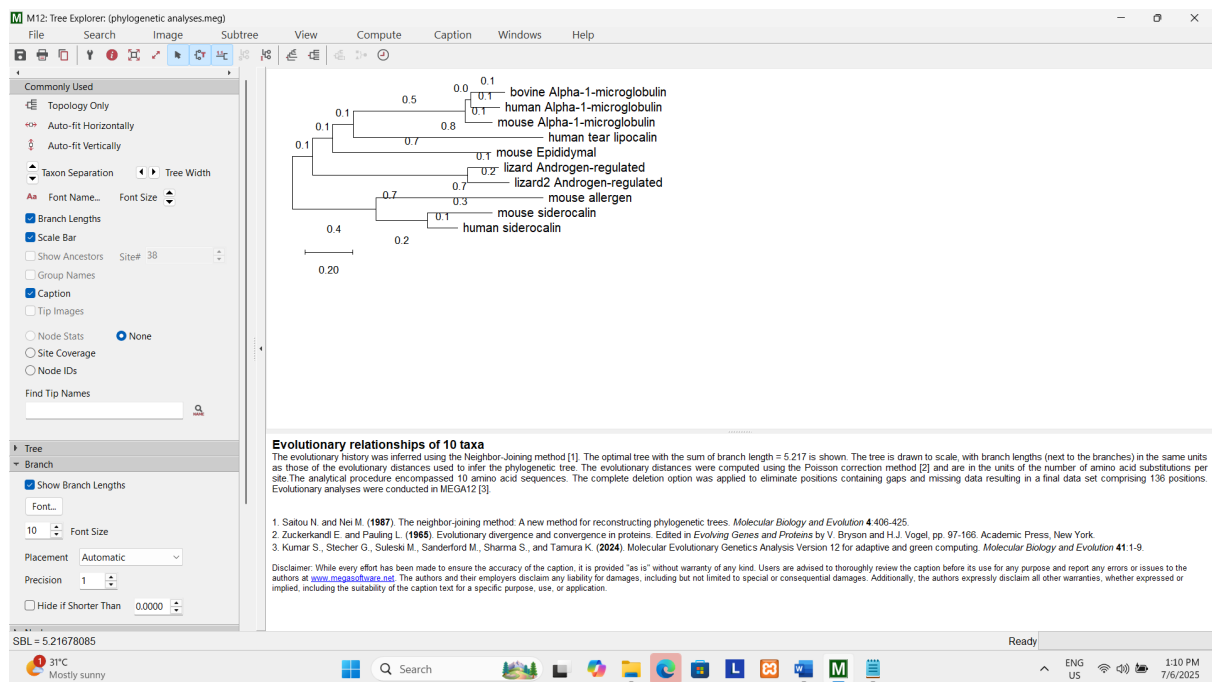
Option	Setting
ANALYSIS	
Scope	<i>All Selected Taxa</i>
Statistical Method →	<i>Neighbor-joining</i>
PHYLOGENY TEST	
Test of Phylogeny →	<i>None</i>
SUBSTITUTION MODEL	
Substitutions Type →	<i>Amino acid</i>
Model/Method →	<i>Poisson model</i>
RATES AND PATTERNS	
Rates among Sites →	<i>Uniform Rates</i>
Pattern among Lineages →	<i>Same (Homogeneous)</i>
DATA SUBSET TO USE	
Gaps/Missing Data →	<i>Complete deletion</i>

Help

Cancel

Ok

The results of a neighbor-joining tree made in MEGA which includes an option to display the caption:



Next, choose Phylogeny (top menu) and select the method maximum likelihood of tree construction :

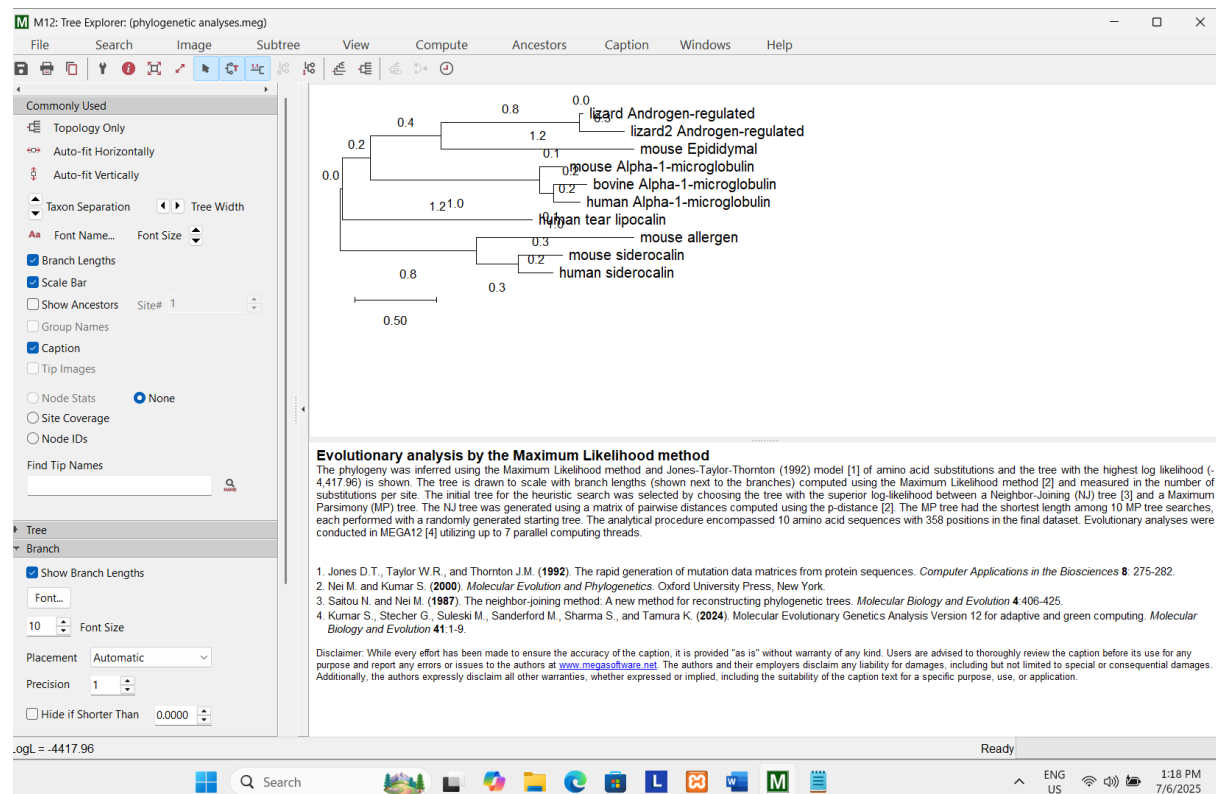
M12: Analysis Preferences

Phylogeny Reconstruction

Option	Setting
ANALYSIS	
Statistical Method →	Maximum Likelihood
PHYLOGENY TEST	
Test of Phylogeny →	None
SUBSTITUTION MODEL	
Substitutions Type →	Amino acid
Model/Method →	Jones-Taylor-Thornton (JTT) model
RATES AND PATTERNS	
Rates among Sites →	Uniform Rates
DATA SUBSET TO USE	
Gaps/Missing Data →	Use all sites
TREE INFERENCE OPTIONS	
ML Heuristic Method →	Nearest-Neighbor-Interchange (NNI)
Initial Tree for ML →	Make initial tree automatically (Default - NJ/MP)
Branch Swap Filter →	None
SYSTEM RESOURCE USAGE	
Number of Threads →	7

Help Cancel Ok

The results of a maximum likelihood tree made in MEGA which includes an option to display the caption:



Then, choose Phylogeny (top menu) and select the method Minimum-Evolution of tree construction

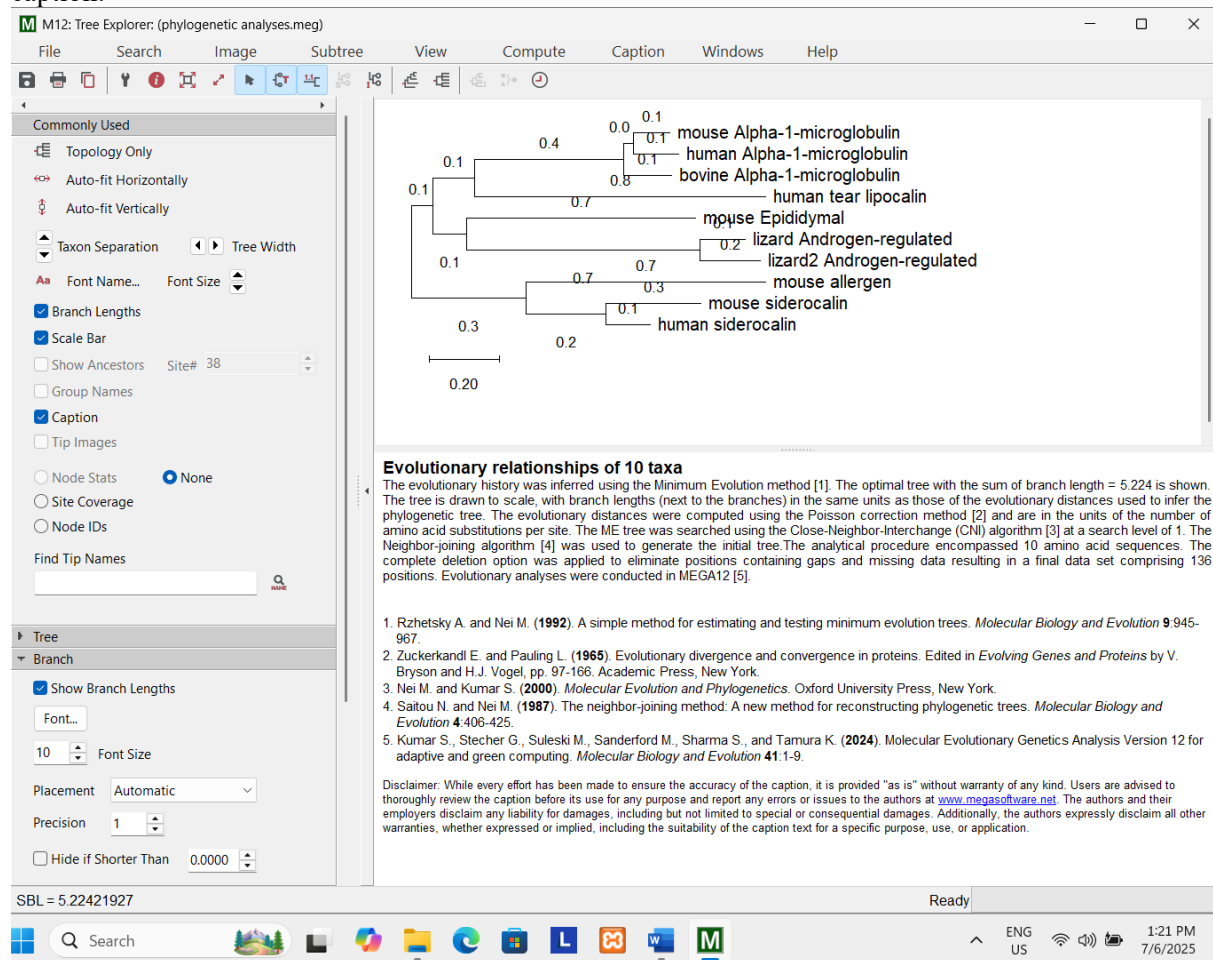
M12: Analysis Preferences

Phylogeny Reconstruction

Option	Setting
ANALYSIS	
Scope	All Selected Taxa
Statistical Method →	Minimum Evolution method
PHYLOGENY TEST	
Test of Phylogeny →	None
SUBSTITUTION MODEL	
Substitutions Type →	Amino acid
Model/Method →	Poisson model
RATES AND PATTERNS	
Rates among Sites →	Uniform Rates
Pattern among Lineages →	Same (Homogeneous)
DATA SUBSET TO USE	
Gaps/Missing Data →	Complete deletion
TREE INFERENCE OPTIONS	
ME Heuristic Method →	Close-Neighbor-Interchange (CNI)
Initial Tree for ME →	Obtain initial tree by Neighbor-Joining
ME Search Level →	1

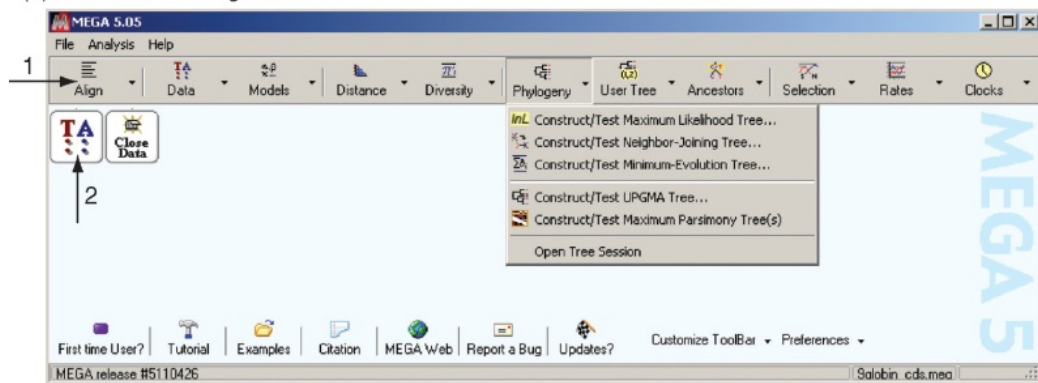
Buttons: ? Help, X Cancel, ✓ Ok

The results of a Minimum-Evolution tree made in MEGA which includes an option to display the caption:

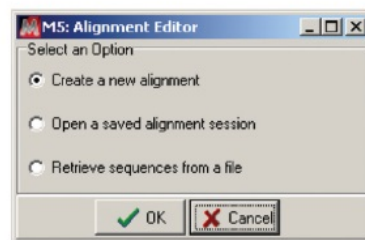


- viii) Clades with Low Bootstrap Support that shown from the tress, the grouping of lizard androgen-regulated proteins, some nodes show values around 0.2-0.3. Besides, the separation between mouse allergen and siderocalin sequences, its bootstrap values appear to be around 0.1-0.3. Also, several branches show moderate to low support values in some internal nodes in the Alpha-1-microglobulin cluster. The low bootstrap support in these clades occurs due to several factors likes limited phylogenetic signal, rapid evolutionary divergence, sequence alignment issues, functional constraints and conflicting evolutionary signals.

(a) MEGA main dialog box



(b) Alignment editor



(c) Analysis preferences (for making a neighbor-joining tree)

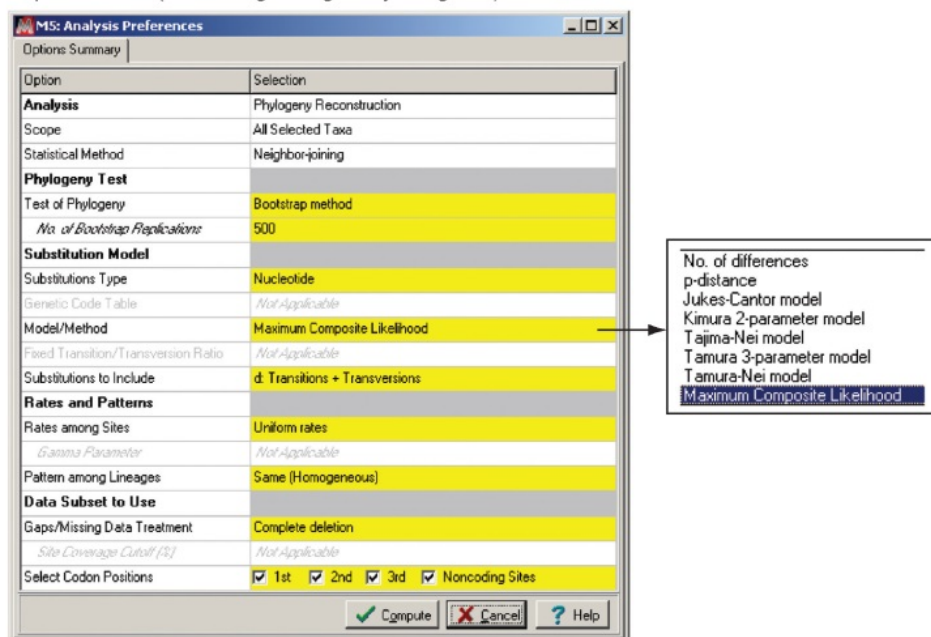


FIGURE 7.9 Using MEGA to make and analyze phylogenetic trees. (a) The main dialog box includes an option to import and align DNA or protein sequences (arrow 1). Once entered the data can be viewed (arrow 2) and manipulated (e.g., you may include/exclude particular taxa or sequence positions). The pull-down menu for phylogenetic analysis is shown. (b) Alignment editor. It is good practice to save sequences for analysis as a text file so they can be studied later. (c) Analysis options for making a neighbor-joining tree. Similar preference boxes are available for other tree-making and analysis methods. Options for the Model/Method are shown to the right.

Bioinformatics and Functional Genomics, Third Edition, Jonathan Pevsner.
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 Companion Website: www.wiley.com/go/pevsnerbioinformatics

Figure 7.9: Pevsner book about using MEGA